

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1068

Country Club 3 100 kV
January 16, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1068. The Interconnection Customer proposes a 145.2 MW solar photovoltaic facility interconnecting at the Country Club 3 100 kV substation in the Indiana area. The request was received on September 26, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 384 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$44,539,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Powerline Elkhorn / Websterville 100 KV
Contingency: Montpelier Electric Generating Sta 100/1 kV Auto
Loading: 159.9% (Rating: 134 MVA)
DFAX: 6.7%
Pre-project loading: 152.6%
2. Facility: Powerline Elkhorn / Thousand Springs 100 KV
Contingency: Montpelier Electric Generating Sta 100/1kV TX
Loading: 154.9% (Rating: 127 MVA)
DFAX: 6.7%
Pre-project loading: 147.2%
3. Facility: Eldean / Monashee 100 KV Line
Contingency: Montpelier Electric Generating Station 100/1 kV Transformer
Loading: 144.4% (Rating: 136 MVA)
DFAX: 6.4%
Pre-project loading: 137.6%
4. Facility: West Asheville / Websterville 100 KV line
Contingency: Muskogee / Montpelier Electric Generating Station 500 KV Line
Loading: 136.9% (Rating: 125 MVA)
DFAX: 5.3%
Pre-project loading: 130.7%
5. Facility: Caguanas 8103 / Columbia Hills Substation 100 KV L
Contingency: Montpelier Electric Generating Sta 100/1 kV Autotransformer
Loading: 134.8% (Rating: 89 MVA)
DFAX: 5.9%
Pre-project loading: 125.2%

6. Facility: Carbondale Nw / Eldean 100 KV
Contingency: Montpelier Electric Generating Sta 100 kV/1 kV Transformer Bank
Loading: 130.4% (Rating: 165 MVA)
DFAX: 6.4%
Pre-project loading: 124.8%
7. Facility: Coso 7 / Montpelier Electric Generating Sta 100 KV
Contingency: Muskogee / Montpelier Electric Generating Station 500 KV Line
Loading: 128.5% (Rating: 156 MVA)
DFAX: 14.7%
Pre-project loading: 114.8%
8. Facility: Richard Lake / Carbondale Nw 100 KV
Contingency: Montpelier Electric Generating Station 100 kV/1 kV Autotransformer
Loading: 124.4% (Rating: 192 MVA)
DFAX: 6.4%
Pre-project loading: 119.5%
9. Facility: Muskogee 500 kV/230 kV XFMR
Contingency: Muskogee / Swan Vly 500 KV
Loading: 121.9% (Rating: 1,410 MVA)
DFAX: 16.3%
Pre-project loading: 120.2%
10. Facility: W Melrose / Trego 230 KV Line
Contingency: Swan Vly / Trego 500 KV
Loading: 121.2% (Rating: 623 MVA)
DFAX: 6.4%
Pre-project loading: 119.7%
11. Facility: Richard Lake Substation / Savona 100 KV
Contingency: Montpelier Electric Generating Sta 100-1 kV Transformer Bank
Loading: 119.8% (Rating: 219 MVA)
DFAX: 6.4%
Pre-project loading: 115.5%
12. Facility: Monashee / W Riverside 100 KV
Contingency: Montpelier Electric Generating Sta 100 kV/1 kV Transformer
Loading: 118.5% (Rating: 145 MVA)
DFAX: 6.4%
Pre-project loading: 112.1%
13. Facility: Timber Switch / Muskogee 230 KV
Contingency: Swan Vly / Trego 500 KV
Loading: 118.5% (Rating: 731 MVA)
DFAX: 6.5%
Pre-project loading: 117.2%
14. Facility: Caguanas 8103 / Sheepskin 100 KV
Contingency: Montpelier Electric Generating Station 100 kV/1 kV Transformer Bank
Loading: 118.0% (Rating: 109 MVA)
DFAX: 5.9%
Pre-project loading: 110.1%
15. Facility: Butler Municipal Power Plant Substation / Savona Substation 100 KV
Contingency: Montpelier Electric Generating Station 100/1kV Auto
Loading: 117.3% (Rating: 237 MVA)
DFAX: 6.4%
Pre-project loading: 113.3%
16. Facility: 15Th Avenue S / Norwich 230 KV
Contingency: Omni-Hackberry / Norwich 500 KV

- Loading: 116.5% (Rating: 939 MVA)
DFAX: 5.6%
Pre-project loading: 115.6%
17. Facility: Timber Switch / W elrose 230 KV
Contingency: Swan Vly / Trego 500 KV
Loading: 116.2% (Rating: 770 MVA)
DFAX: 6.4%
Pre-project loading: 115.0%
18. Facility: Hansel / Butler Municipal Power Plant 100 KV Line
Contingency: Montpelier Electric Generating Station 100/1kV TX
Loading: 115.5% (Rating: 253 MVA)
DFAX: 6.4%
Pre-project loading: 111.8%
19. Facility: Dutton Pump / Leyendecker 100 KV
Contingency: Montpelier Electric Generating Sta 100/1 kV Autotransformer
Loading: 114.5% (Rating: 144 MVA)
DFAX: 6.0%
Pre-project loading: 108.5%
20. Facility: Hansel / Thousand Sprins 100 KV line
Contingency: Montpelier Electric Generating Sta 100/1kV XFMR
Loading: 114.3% (Rating: 263 MVA)
DFAX: 6.4%
Pre-project loading: 110.8%
21. Facility: Azle / Country Club 3 100 KV L
Contingency: Signetics / Country Club 3 100 KV Line
Loading: 108.5% (Rating: 124 MVA)
DFAX: 100.0%
Pre-project loading: 8.9%
22. Facility: Lonetree 500/230kV Auto
Contingency: E Methuen / Lonetree 500 KV Line
Loading: 108.5% (Rating: 1,165 MVA)
DFAX: 6.5%
Pre-project loading: 107.6%
23. Facility: Henry County 500/230kV Autotransformer
Contingency: Sabanera / Lonetree 500 KV
Loading: 106.9% (Rating: 1,903 MVA)
DFAX: 12.6%
Pre-project loading: 106.0%
24. Facility: Signetics / Country Club 3 100 KV
Contingency: Country Club 3 / Azle 100 KV
Loading: 103.6% (Rating: 129 MVA)
DFAX: 100.0%
Pre-project loading: 8.5%
25. Facility: Eastown Road / Dean Forest 230 KV
Contingency: Swan Vly / Muskogee 500 KV
Loading: 102.2% (Rating: 1,128 MVA)
DFAX: 9.7%
Pre-project loading: 101.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Powerline Elkhorn / Websterville 100 KV	Rebuild line with higher-capacity conductor	\$11,412,000	31% (\$3,496,000)
Thousand Springs / Powerline Elkhorn 100 KV Line	Rebuild line with higher-capacity conductor	\$2,169,000	31% (\$664,000)
Eldean / Monashee 100 KV	Rebuild line with higher-capacity conductor	\$3,028,000	22% (\$653,000)
West Asheville / Websterville 100 KV	Rebuild line with higher-capacity conductor	\$30,006,000	22% (\$6,652,000)
Caguanas 8103 Substation / Columbia Hills 100 KV	Rebuild line with higher-capacity conductor	\$5,003,000	55% (\$2,753,000)
Carbondale Nw / Eldean 100 KV	Rebuild line with higher-capacity conductor	\$3,906,000	22% (\$842,000)
Coso 7 / Montpelier Electric Generating Sta 100 KV	Rebuild line with higher-capacity conductor	\$17,826,000	48% (\$8,535,000)
Richard Lake / Carbondale Nw 100 KV line	Rebuild line with higher-capacity conductor	\$2,833,000	22% (\$611,000)
Muskogee 500 kV/230 kV Auto	Replace transformer with higher-rated unit	\$24,569,000	3% (\$669,000)
W Melrose / Trego 230 KV	Rebuild line with higher-capacity conductor	\$9,986,000	2% (\$194,000)
Richard Lake / Savona Substation 100 KV Line	Rebuild line with higher-capacity conductor	\$6,138,000	22% (\$1,323,000)
Monashee / West Riverside 100 KV	Rebuild line with higher-capacity conductor	\$4,293,000	22% (\$925,000)
Timber Switch / Muskogee 230 KV	Rebuild line with higher-capacity conductor	\$18,322,000	2% (\$357,000)
Caguanas 8103 / Sheepskin 100 KV	Rebuild line with higher-capacity conductor	\$6,538,000	55% (\$3,597,000)
Butler Municipal Power Plant Substation / Savona Substation 100 KV	Rebuild line with higher-capacity conductor	\$1,522,000	22% (\$328,000)
15Th Avenue South / Norich 230 KV L	Rebuild line with higher-capacity conductor	\$17,813,000	1% (\$215,000)
Timber Switch / W elrose 230 KV	Rebuild line with higher-capacity conductor	\$91,944,000	2% (\$1,803,000)
Hansel / Butler Municipal Power Plant 100 KV Line	Rebuild line with higher-capacity conductor	\$5,948,000	22% (\$1,282,000)
Dutton Pump / Leyendecker 100 KV	Rebuild line with higher-capacity conductor	\$9,980,000	27% (\$2,666,000)
Hansel / Thousand Sprins 100 KV line	Rebuild line with higher-capacity conductor	\$5,902,000	22% (\$1,272,000)
Azle / Country Club 3 100 KV L	Reconductor line	\$3,769,000	100% (\$3,769,000)
Lonetree 500/230kV Auto	Replace transformer with higher-rated unit	\$27,300,000	2% (\$480,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Henry Couty 500-230 kV Transformer	Replace transformer with higher-rated unit	\$30,152,000	1% (\$322,000)
Signetics / Country Club 3 100 KV	Reconductor line	\$992,000	100% (\$992,000)
Dean Forest / Eastown Road 230 KV line	Reconductor line	\$5,968,000	2% (\$139,000)
	Total allocated to project		\$44,539,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.